M201103W- Signals and Systems

ssch7_6 –Description of D-T LTI systems in the z- domain

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

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Description of D-T LTI systems in the z- domain

※ System function of D-T LTI systems

If the difference equation describing a D-T LTI system is:

$$a_n y[k-n]+\cdots+a_1 y[k-1]+a_0 y[k]= b_m x[k-m]+\cdots+b_1 x[k-1]+b_0 x[k]$$

Do the z -transform using shifting property, we get:

$$a_n z^{-n} Y_{zs}(z)+\cdots+a_1 z^{-1} Y_{zs}(z)+a_0 Y_{zs}(z)= b_m z^{-m} X(z)+\cdots+b_1 z^{-1} X(z)+b_0 X(z)$$

Definition of system function:

$$H(z)=\frac{Y_{zs}(z)}{X(z)}=\frac{b_m z^{-m}+b_{m-1} z^{-m+1}+\cdots+b_1 z^{-1}+b_0}{a_n z^{-n}+a_{n-1} z^{-n+1}+\cdots+a_1 z^{-1}+a_0}$$

Description of D-T LTI systems in the z- domain

► Relation between system function $H(z)$ and impulse response $h[k]$.

$$\delta[k] \longrightarrow \boxed{\text{D-T LTI}} \longrightarrow h[k]$$

$$H(z) = \frac{Y_{zs}(z)}{X(z)} = \frac{\mathcal{Z}\{h[k]\}}{\mathcal{Z}\{\delta[k]\}} = \mathcal{Z}\{h[k]\}$$

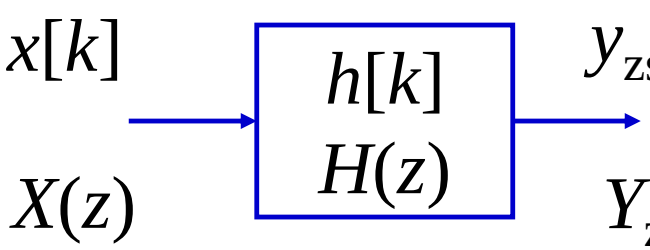
$$h[k] = \mathcal{Z}^{-1}\{H(z)\}$$

Summary:

$H(z)$ and $h[k]$ are a pair of z-transform.

Description of D-T LTI systems in the z- domain

- Relation between $H(z)$ and $Y_{zs}(z)$.



$$y_{zs}[k] = x[k]*h[k]$$

$$Y_{zs}(z) = X(z)H(z)$$

Description of D-T LTI systems in the z- domain

► Several methods to solve $H(z)$.

① From $h[k]$: $H(z)=\mathscr{Z}\{h[k]\}$

② From output and input $H(z)=\dfrac{\mathscr{Z}\{y_{zs}[k]\}}{\mathscr{Z}\{x[k]\}}$

③ From difference equation

Example. Find the system function $H(z)$ of unit delay $y[k] = x[k-1]$.

Solution:

$$h[k] = \delta[k-1]$$

$H(z)$ and $h[k]$ are a pair of z- transform.

$$H(z) = \mathscr{Z}\{h[k]\}$$

$H(z)$ of unit delay is:

$$H(z) = z^{-1}$$

Example. In a D-T LTI system, if output is $y_{zs}[k] = (0.5)^k u[k]+4u[k]$ when input is $x[k]=u[k]$. Calculate $H(z)$.

Solution: Do z-transform both on $x[k]$ and $y_{zs}[k]$, we get:

$$X(z) = \mathscr{Z}\{x[k]\} = \mathscr{Z}\{u[k]\} = \frac{z}{z-1}, \qquad |z|>1$$

$$Y_{zs}(z) = \mathscr{Z}\{y_{zs}[k]\} = \mathscr{Z}\{0.5^k u[k]+4u[k]\}$$

$$= \frac{z}{z-0.5} + \frac{4z}{z-1} = \frac{5z^2-3z}{(z-0.5)(z-1)}, \qquad |z|>1$$

According to the definition of $H(z)$,

$$H(z) = \frac{\mathscr{Z}\{y_{zs}[k]\}}{\mathscr{Z}\{x[k]\}} = \frac{Y_{zs}(z)}{X(z)} = \frac{5z-3}{z-0.5}, \qquad |z|>0.5$$

Example. Consider the difference equation of a D-T causal LTI is below.

$y[k]+3y[k-1]+2y[k-2]=3x[k]-x[k-1]$. Calculate $H(z)$.

Solution: Do z-transform on the difference equation, we get:

$$Y_{zs}(z)+3z^{-1}Y_{zs}(z)+2z^{-2}Y_{zs}(z)=3X(z)-z^{-1}X(z)$$

According to the definition of $H(z)$,

$$H(z)=\frac{Y_{zs}(z)}{X(z)}=\frac{3-z^{-1}}{1+3z^{-1}+2z^{-2}}$$

$$=\frac{3z^2-z}{z^2+3z+2},\quad |z|>2$$